

APPLIED MATHEMATICS: DYNAMICS

Official Ministry of Education Textbook - Grade 12

ACADEMIC YEAR: 2025 / 2026 | GRADE 12 (GENERAL SECONDARY CERTIFICATE)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

The authorized Ministry textbook for Dynamics, covering variable acceleration kinematics, Newtons three laws, multi-body pulley configurations, impact mechanics, and conservation of mechanical energy.

CORE CURRICULUM HIGHLIGHTS:

- Variable acceleration kinematics with $a = v(dv/dx)$
- Newtons 2nd Law applied to accelerating elevators and pulleys on inclined planes
- Impulse as area under $F-t$ curve and conservation of linear momentum
- Work-energy theorem, conservative fields, and instantaneous engine power

APPROVED CHAPTER MODULES:

- Rectilinear Kinematics & Differentiation of Vector Functions	pp. 1 54 (3 core units)
- Newton's Laws of Motion, Pulleys & Connected Bodies	pp. 55 118 (4 core units)
- Impulse and Direct Linear Collisions	pp. 119 162 (2 core units)
- Work, Kinetic Energy, Potential Energy & Mechanical Power	pp. 163 216 (4 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Official Concept & Formula Compendium (8p - 21.5 KB)
Full Ministerial Textbook: 216 Pages - 19.5 MB (Direct on MOE Portal)
Publisher: Ministry of Education & Technical Education - Arab Republic of Egypt
Official Portal: <https://moe.gov.eg/ar/elearning-content/> - Code: TH-DYN-G12
Legal Deposit: 2025/14892 - ISBN: 978-977-12-8135-98



MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

The authorized Ministry textbook for Dynamics, covering variable acceleration kinematics, Newtons three laws, multi-body pulley configurations, impact mechanics, and conservation of mechanical energy. Variable acceleration kinematics with $a = v(dv/dx)$ Newtons 2nd Law applied to accelerating elevators and pulleys on inclined planes Impulse as area under F-t curve and conservation of linear momentum Work-energy theorem, conservative fields, and instantaneous engine power

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	40% Understanding & Application Calculations, proofs & models	30% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal:

<https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Rectilinear Kinematics & Differentiation of Vector Functions	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Newton's Laws of Motion, Pulleys & Connected Bodies	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Impulse and Direct Linear Collisions	7 wks	15 pts	25.0%	2 core instructional units covering national standards and
Work, Kinetic Energy, Potential Energy & Mechanical Power	7 wks	15 pts	25.0%	4 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism)

Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.

Engineering Sciences (Civil, Mechanical, Electrical)

Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.

Computer Science & Algorithmics

Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: RECTILINEAR KINEMATICS & DIFFERENTIATION OF VECTOR FUNCTIONS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Rectilinear Kinematics & Differentiation of Vector Functions.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Rectilinear Kinematics & Differentiation of Vector Functions.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: NEWTON'S LAWS OF MOTION, PULLEYS & CONNECTED BODIES

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Newton's Laws of Motion, Pulleys & Connected Bodies.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Newton's Laws of Motion, Pulleys & Connected Bodies.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: IMPULSE AND DIRECT LINEAR COLLISIONS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Impulse and Direct Linear Collisions.
- [+] Synthesize and apply analytical frameworks across all 2 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Impulse and Direct Linear Collisions.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

APPLIED MATHEMATICS: DYNAMICS - CORE PRINCIPLES & AXIOMS
Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion: $\sum F_{net} = 0, \sum M_{pivot} = 0$
Continuity & Conservation Standard: $\text{div}(\mathbf{J}) + \frac{\partial \rho}{\partial t} = 0$
Normalized Standard Score: $Z = \frac{(X - \mu)}{\sigma}$

PROCEDURAL ASSESSMENT & TRANSFORMATION RULES
Linear Operator Transformation: $T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$
Sample Variance Boundary: $s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio: $\eta = (\text{Output}_{actual} / \text{Input}_{standard}) \cdot 100\%$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

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